Indoor Location Prediction from Wi-Fi Signal Strength: A Machine Learning Baseline on UJIIndoorLoc

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Abstract—GPS does not work reliably indoors, but almost every building is covered by Wi-Fi. We predict a phone's building, floor and (x, y) position from a single scan of Wi-Fi received signal strengths (RSSI), using the public UJIIndoorLoc dataset (3 buildings, 13 building–floor classes, 520 access points, 21,048 scans). The data is extremely sparse: 96.5% of the readings are "not detected". We clean the data, handle the missing-signal convention consistently, and compare linear/logistic regression, k-nearest neighbours and Random Forests, including a hierarchical Random Forest that first predicts the building and then uses a per-building model. All choices are made by 5-fold cross-validation grouped by user, and the official validation file, collected four months later, is used only once as the test set. On the test set, the per-building Random Forest reaches 99.8% building accuracy, 91.3% floor accuracy and 9.37 m mean position error (7.07 m median). k-NN achieves the lowest mean error, 8.69 m, and linear regression fails at positioning (52 m), which shows that the signal-to-position mapping is strongly non-linear. We analyse the errors by building/floor correctness, by phone and by signal coverage, and identify the gap that Phase 2 (multitask deep learning) will address.

Index Terms—indoor positioning, Wi-Fi fingerprinting, RSSI, random forest, k-nearest neighbours, UJIIndoorLoc

I. Introduction

Satellite positioning fails inside buildings because walls and roofs block and reflect the signal. Indoor location is still useful for navigation in campuses and hospitals, for emergency response and for asset tracking. Wi-Fi access points (APs) are already everywhere, and a phone can measure the RSSI of every AP it hears without connecting to it. Converting RSSI to distance with a propagation model is unreliable indoors because of walls, people and multipath reflections [1]. Fingerprinting avoids modelling propagation: a radio map of RSSI vectors is recorded at known positions, and a new scan is located by learning from that map [2].

The task combines two classification problems (building, floor) and one regression problem (x, y in metres). Phase 1 of this project builds and validates the machine learning baseline. Our contributions are:

- a careful data audit (duplicates, empty scans, a faulty phone, train/test shift);
- a leakage-free pipeline in which every statistic is fitted on training data, and model selection uses cross-validation that holds out whole users;
- a comparison of linear/logistic regression, k-NN and Random Forests (plain and hierarchical) on building accuracy, floor accuracy, coordinate error and the IPIN competition score;
- a hierarchical error breakdown, floor confusion matrices, coordinate error maps and a per-phone robustness analysis.

II. Literature Review

Fingerprinting. RADAR [2] introduced Wi-Fi fingerprinting with nearest-neighbour matching in signal space. Horus [3] modelled the RSSI distribution at each location probabilistically, which showed how noisy single readings are.

UJIIndoorLoc. Torres-Sospedra et al. [4] released our dataset, the first large multi-building, multi-floor public benchmark, with a k-NN baseline reaching 89.9% floor accuracy and 7.9 m mean error. A follow-up study [5] compared 51 distance measures and four data representations for k-NN and found that the powed representation, which compresses weak readings, helps. The IPIN 2016 competition [6] defined the combined score we report.

Machine learning models. Bozkurt et al. [7] compared decision trees, naive Bayes, k-NN and ensembles on UJIIndoorLoc and found k-NN and tree ensembles strongest. HybLoc [8] splits the radio map into clusters and trains a Random Forest [9] per cluster, the same divide-and-conquer idea as our per-building model.

Deep learning (Phase 2 context). Stacked autoencoders with an MLP [10], [11] and CNNs [12] reach about 92–96% floor accuracy on UJIIndoorLoc. Multitask learning [13], where related tasks share one representation, motivates our Phase 2 model.

Gaps addressed here. (i) The "+100 = not detected" value is often fed to models as a number, which inverts its meaning. (ii) Random train/validation splits leak near-duplicate scans and overstate accuracy. (iii) Most baselines model building, floor and position independently.

III. Dataset and Exploratory Analysis

UJIIndoorLoc [4] was collected on the Universitat Jaume I campus: three buildings with four, four and five floors. Each scan contains 520 RSSI values (dBm; +100

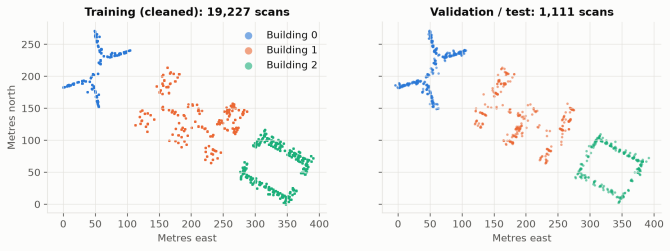


Fig. 1. Scan positions. The test set covers the same buildings but is skewed towards building 0.

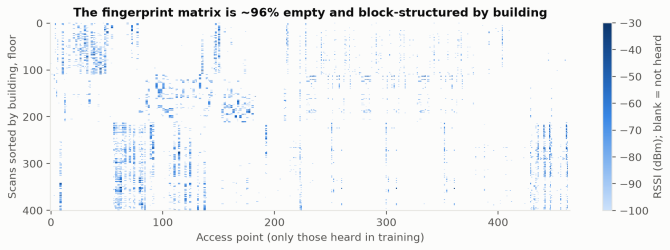


Fig. 2. Sample of the fingerprint matrix (rows sorted by building and floor). It is about 96% empty, with block structure per building.

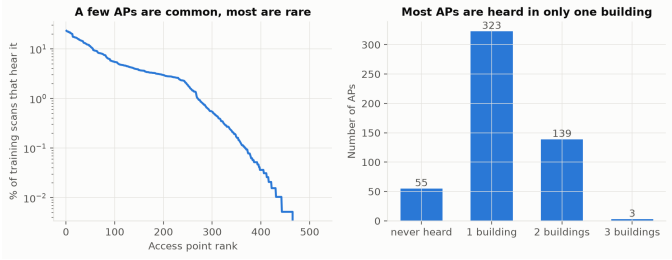


Fig. 3. Left: a few APs are heard often, most rarely. Right: most APs are heard in a single building.

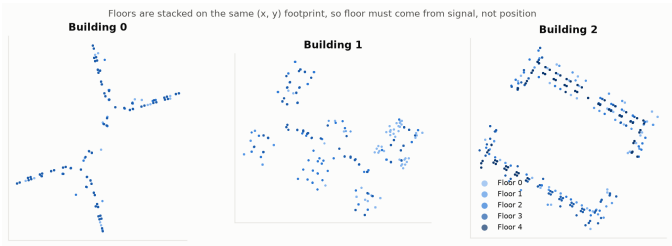


Fig. 4. Training positions coloured by floor. Floors are stacked on the same (x, y) footprint.

means not detected) plus longitude and latitude in metres (UTM), floor, building, space ID, user ID, phone ID and a timestamp. The training file has 19,937 scans from 18 users and 16 phones (May–June 2013). The validation file has 1,111 scans collected four months later (Sept–Oct 2013) by different users, on 11 phones, 9 of which do not appear in training.

Cleaning. We removed 637 exact duplicate rows and 73 scans in which no AP was heard, leaving 19,227 training scans. The test file is left untouched so that our results stay comparable with the literature.

Sparsity. A scan hears about 18 of the 520 APs on average, so 96.5% of the matrix is "not detected" (Fig. 2). Detected RSSI ranges from -104 to 0 dBm, and most readings are weaker than -70 dBm. 55 APs are never heard in training, and 323 of the remaining 465 are heard in only one building (Fig. 3). This is why building classification is easy.

Floors. Floors share the same footprint (Fig. 4), so position cannot reveal the floor; it must come from which APs are heard and how strongly.

Class balance and shift. Training is fairly balanced over the 13 building–floor classes. The test set is skewed: 48% of test scans are in building 0, compared with 27% in training (Fig. 5). Test phone proportions also differ (Fig. 6), and 64% of test scans come from phones never seen in training. This makes the test set a realistic new users, new devices, later date evaluation.

Data quality. 492 training scans contain readings above -30 dBm, which is implausibly strong for indoor Wi-Fi. 430 of them come from user 6 with phone 19 (Fig. 7). We keep them by default and test the effect of removing them in Section V.

PCA. With the missing values handled, the first two principal components already separate the three buildings (Fig. 8), but 50 components explain only 84% of the variance. The useful information is spread over many APs.

IV. Methodology

A. Problem formulation

Let $\mathbf{x} \in \mathbb{R}^{520}$ be a scan. We learn $f_b(\mathbf{x}) \in \{0, 1, 2\}$, $f_f(\mathbf{x}) \in \{0, \ldots, 4\}$ and $f_p(\mathbf{x}) \in \mathbb{R}^2$. Following the IPIN competition [6], the combined score of one prediction is

$$s = \|\mathbf{p} - \hat{\mathbf{p}}\|_2 + 50 \cdot [b \neq \hat{b}] + 4 \cdot [f \neq \hat{f}], \qquad (1)$$

so a wrong building costs as much as 50 m of error and a wrong floor costs 4 m.

B. Feature engineering

Missing signals. "Not detected" means the signal is below the phone's sensitivity, so it is missing not at random: its absence tells us the AP is far away. Used as a number, +100 dBm would look like the strongest possible signal. We replace it with $m = \min(\text{observed}) - 1 = -105$ dBm, computed on training data only. APs never heard in training are dropped, which leaves 465 features.

Variants compared. (a) raw: dBm with -105 for not detected; (b) positive: $x - m$, so not detected is 0; (c) powed: $\big((x - m)/(x_{\max} - m)\big)^e$, which compresses weak readings [5]; (d) raw plus one binary heard/not-heard flag per AP; (e) (d) plus per-scan summaries (number of APs heard, strongest, mean and standard deviation of RSSI, and the IDs of the three strongest APs). The variant for each model is chosen by cross-validation.

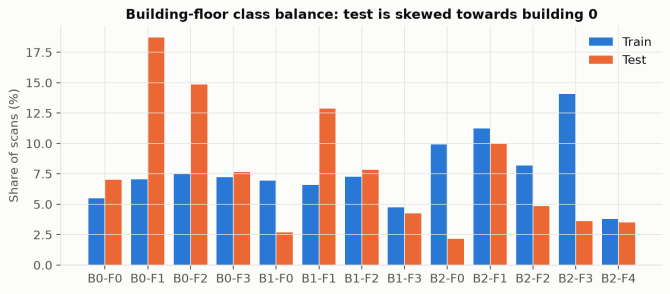


Fig. 5. Share of scans per building–floor in train and test.

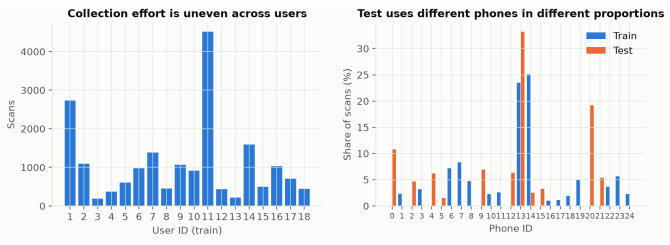


Fig. 6. Scans per user (train) and per phone (train vs. test).

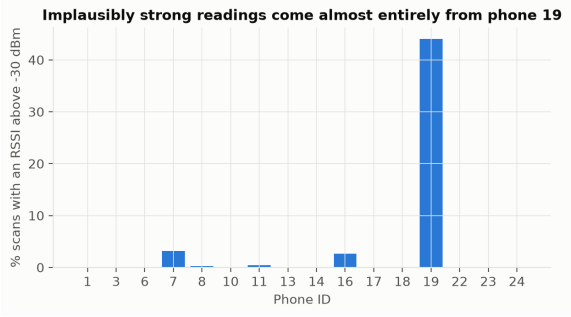


Fig. 7. Share of scans with an RSSI above −30 dBm, by phone.

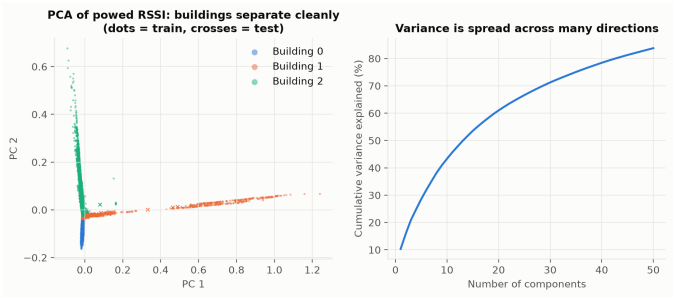


Fig. 8. PCA of the RSSI matrix. Buildings separate; variance is spread across many directions.

C. Models

Centroid baseline. Predicts the most common building and floor and the mean position. It gives a floor against which to measure the other models.

Linear and logistic regression. Inputs are standardised with training statistics. Logistic regression (softmax, cross-entropy, L2, $C = 1$) predicts building and floor. Ordinary least-squares linear regression predicts x and y.

k-NN [14]. The classic fingerprinting method: find the k training scans nearest in Euclidean distance. Building and floor come from a distance-weighted vote, and the position is the distance-weighted mean of the neighbours' positions.

Random Forest [9]. An ensemble of CART trees. Each tree is grown on a bootstrap sample, and each split considers a random subset of max_features APs. Splits minimise Gini impurity $G = 1 - \sum_k p_k^2$ for classification and variance for regression. For B trees with variance σ^2 and pairwise correlation ρ, the variance of their average is

$$\mathrm{Var}\left(\frac{1}{B}\sum_b T_b\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2, \qquad (2)$$

so bagging removes the second term as B grows and random feature selection lowers ρ. Because a tree only compares a feature with a threshold, any order-preserving transform of the input gives the same tree. A regression leaf predicts the mean of real training positions, so predictions never leave the surveyed area.

Plain vs. per-building (hierarchical). The plain variant trains three independent forests on all scans, as in the project brief. The per-building variant predicts the building first, then sends the scan to a floor forest and a position forest trained only on that building's scans. This turns one global problem into three smaller, more homogeneous ones, at the risk that a wrong building prediction cascades to the floor and position.

D. Validation protocol

Consecutive scans by the same user at the same spot are near-duplicates, so a random split would leak them. We use 5-fold GroupKFold with USERID as the group: no user appears in both the training and validation parts of a fold. Feature variants, k and max_features are all selected by the mean CV IPIN score. Imputation and scaling are re-fitted inside every fold. The official validation file is used once, as the test set, after all choices are fixed.

V. Results

A. Feature comparison

Table I compares the feature variants. The raw and positive variants give identical results for both models, as theory predicts: the shift is order-preserving (trees) and cancels in Euclidean distances (k-NN). None of the richer variants improved the CV score, so the simplest representation (raw dBm with −105 for not detected) is used for every model. Powed features helped k-NN on the test set (IPIN 8.58 vs. 9.22) but hurt it in CV (17.60 vs. 14.46). Because test scores must not drive selection, we did not adopt them.

B. Tuning

Table II shows the CV search. For k-NN, $k = 5$ is best: $k = 1$ is noisy and $k = 11$ over-smooths. For the

TABLE I
Feature comparison. CV = 5-fold GroupKFold by user on the training set (used to choose); test shown for reference only.

Features	CV IPIN	Test IPIN	Test B+F %	Test err. (m)
Random Forest (per-building)				
raw dBm, not-heard = -105	15.23	9.70	91.18	9.31
positive (shifted so not-heard = 0)	15.23	9.70	91.18	9.31
powed (compress weak signals)	15.71	9.88	91.09	9.44
raw + heard/not-heard flags	15.55	10.28	90.28	9.80
raw + flags + scan summary	15.82	9.86	91.00	9.50
k-NN (k=5)				
raw dBm, not-heard = -105	14.46	9.22	90.10	8.69
positive (shifted so not-heard = 0)	14.46	9.22	90.10	8.69
powed (compress weak signals)	17.60	8.58	93.43	8.19
raw + heard/not-heard flags	14.54	9.20	90.01	8.66

TABLE II
Hyper-parameter search with user-grouped 5-fold CV on the training set.

Model	Parameters	B+F %	Mean err. (m)	IPIN
k-NN	k=1	72.38	12.92	15.55
k-NN	k=3	72.30	12.10	14.71
k-NN	k=5	72.29	11.78	14.46
k-NN	k=7	72.26	11.80	14.73
k-NN	k=11	72.06	11.93	15.05
Random Forest	max_features=sqrt, n_estimators=300	70.90	13.55	15.21
Random Forest	max_features=0.1, n_estimators=300	71.56	14.38	16.08
Random Forest	max_features=0.3, n_estimators=300	70.93	15.57	17.45

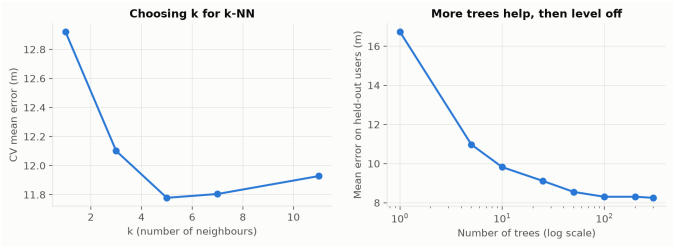


Fig. 9. Left: CV error vs. k. Right: error on held-out users vs. number of trees.

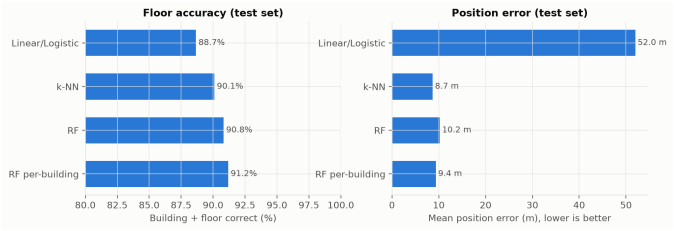


Fig. 10. Floor accuracy and mean position error on the test set (centroid baseline omitted: 3.6%, 142.8 m).

Random Forest, max_features= $\sqrt{p}$ (≈21 APs per split) beats 10% and 30%, because more random, less correlated trees average better. The error drops quickly up to about 50 trees and is flat beyond 100 (Fig. 9), so 300 trees is safely on the plateau. In CV, k-NN scores slightly better than the forest (IPIN 14.46 vs. 15.21; floor accuracy 74.1% vs. 71.3%), while the forest identifies the building more reliably (99.0% vs. 96.7%).

C. Test-set comparison

Table III and Fig. 10 report the single evaluation on the official test set. All learned models identify the building almost perfectly. Logistic regression even gets 100%, because buildings use disjoint APs. On floor, the per-building Random Forest is best (91.3%), ahead of the plain forest (90.9%), k-NN (90.1%) and logistic regression (88.7%). For position, linear regression is unusable (52.0 m), while k-NN (8.69 m) and the per-building forest (9.37 m) are close. The per-building design lowers the forest's error from 10.22 m to 9.37 m and its 95th-percentile error from 27.4 m to 23.3 m, the lowest tail of all models (Fig. 11). For reference, the dataset paper's k-NN baseline reported 89.9% floor accuracy and 7.9 m mean error [4].

D. Error analysis

Hierarchical breakdown (Table IV). For the per-building forest, only 2 of 1,111 scans have the wrong building, but they are far off (107 m). The 96 scans with the right building but wrong floor average 15.7 m, almost twice the 8.6 m of fully correct scans. Most of the total error (83%) still comes from correct-floor scans, so improving within-floor positioning matters as much as floor classification.

Floors (Fig. 12). Building 0 is easy (97.6%). Building 1 is the weak spot (79.8%): 43 of its 143 floor-1 test scans are predicted as floor 2. Errors are almost always between adjacent floors, as expected when signals leak through one ceiling.

Spatial errors (Fig. 13). Building 0 has the lowest error (6.5–7.5 m per floor). The worst building–floors are building 2 floor 2 (17.6 m) and floor 4 (16.2 m).

Coverage and devices (Figs. 14, 15). Scans that hear 5 or fewer APs have the largest error (27.9 m, $n = 18$). Error also rises for scans that hear more than 25 APs. 105 of these 141 scans come from phone 20, a phone absent from training that detects far more APs than the training phones, so its fingerprints look unfamiliar. Across test phones, mean error ranges from 6.2 m (phone 9) to 16.3 m (phone 14), so device heterogeneity is a real source of error.

Outliers and importance. Removing the 492 scans with readings above −30 dBm did not help (9.46 m vs. 9.37 m), so the forest is already robust to them. Feature importance (Fig. 16) shows that the building classifier spreads its importance over many APs instead of relying on a single one.

VI. Discussion

Why linear models fail at position but not at building. Building identity is almost a linear function of which APs are heard, so a weighted sum separates it. Position is not: RSSI falls with the logarithm of distance, walls cut it abruptly, and the same AP is "east of you" in one corridor and "west" in another. A single hyperplane cannot represent this. Trees and k-NN make local, piecewise decisions and therefore can.

k-NN vs. Random Forest. The two are close, and the ranking depends on the evaluation: k-NN is ahead in CV, and on test it has the lower mean (8.69 m) and

TABLE III
Test-set results (official UJIIndoorLoc validation file, 1,111 scans). Accuracies in %, errors in metres. IPIN score = 2-D error + 50× building miss + 4× floor miss.

Model	Building	Floor	Bldg+Floor	Mean err.	Median	P75	IPIN score
Centroid (guess the average)	24.12	15.48	3.60	142.75	167.37	193.37	184.07
Linear + Logistic Regression	100.00	88.66	88.66	51.99	37.53	72.25	52.45
k-NN	99.73	90.10	90.10	8.69	5.35	10.31	9.22
Random Forest	99.82	90.91	90.82	10.22	7.63	13.35	10.68
Random Forest (per-building)	99.82	91.27	91.18	9.37	7.07	12.19	9.81

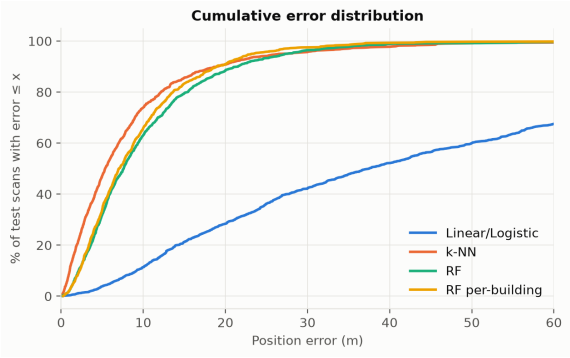


Fig. 11. Cumulative distribution of position error.

TABLE IV
Error split by whether building and floor were right, Random Forest (per-building) (test set).

Case	n	Mean (m)	Median (m)	Share of total error %
building ok, floor ok	1013	8.57	6.65	83.4
building ok, floor wrong	96	15.73	12.55	14.5
building wrong	2	107.25	107.25	2.1

median (5.36 m) error, while the per-building forest has the best floor accuracy and the lightest error tail on test. k-NN copies the positions of real neighbouring fingerprints, which works well when the radio map is dense. The forest learns which APs matter. We keep the Random Forest as the main model because the brief asks for it and it is strongest on floor, the hardest target on the test set, and we report k-NN alongside it.

CV vs. test. CV is much harsher than test (floor accuracy about 72% vs. 91%; mean error 12–14 m vs. 9 m). We ruled out location coverage: 9.9% of CV scans and 7.7% of test scans are more than 5 m from any training point. The likely cause is device novelty. In training, 17 of 18 users carried their own phone, so almost every CV validation scan comes from an unseen device, whereas 35.7% of test scans come from phones seen in training. User-grouped CV is therefore a strict but honest selection criterion.

The gap for Phase 2. All our models treat building, floor and position as separate tasks. The per-building forest links them only through a fixed cascade, and its two wrong-building scans are its worst errors. None of the models learns a representation of the 465 interacting APs that the three tasks could share.

VII. Conclusion and Phase 2 Plan

We built a leakage-free machine learning pipeline for Wi-Fi indoor positioning on UJIIndoorLoc. The key data decisions were to treat "not detected" as a very weak signal (−105 dBm) fitted on training data only, and to validate by holding out whole users. Simple raw features were as good as the engineered variants. A per-building Random Forest reaches 99.8% building accuracy, 91.3% floor accuracy and 9.37 m mean error on a test set collected months later with mostly unseen phones. k-NN is a close second. Remaining errors come from adjacent-floor confusion (especially building 1), scans that hear very few APs, and device differences.

In Phase 2 we will train a multitask MLP with a shared trunk and three heads (building and floor softmax, coordinate regression), optimised with $\mathcal{L} = \lambda_b \mathrm{CE}_b + \lambda_f \mathrm{CE}_f + \lambda_p \mathrm{MSE}_p$ [13], regularised with dropout, weight decay and early stopping on the same user-grouped validation. We will compare it with these baselines on the identical split and metrics, and run ablations (single-task vs. multitask, loss weights).

Team Contributions

Member 1	Literature review, problem framing, evaluation metrics
Member 2	Data cleaning and exploratory data analysis
Member 3	Feature engineering and feature comparison
Member 4	Linear/logistic and k-NN models, cross-validation design
Member 5	Random Forest models, error analysis, repository and report

Reproducibility

Code, figures and an executed notebook are in the project GitHub repository. Running python scripts/download_data.py, run_eda.py and run_experiments.py reproduces every number (random seed 42).

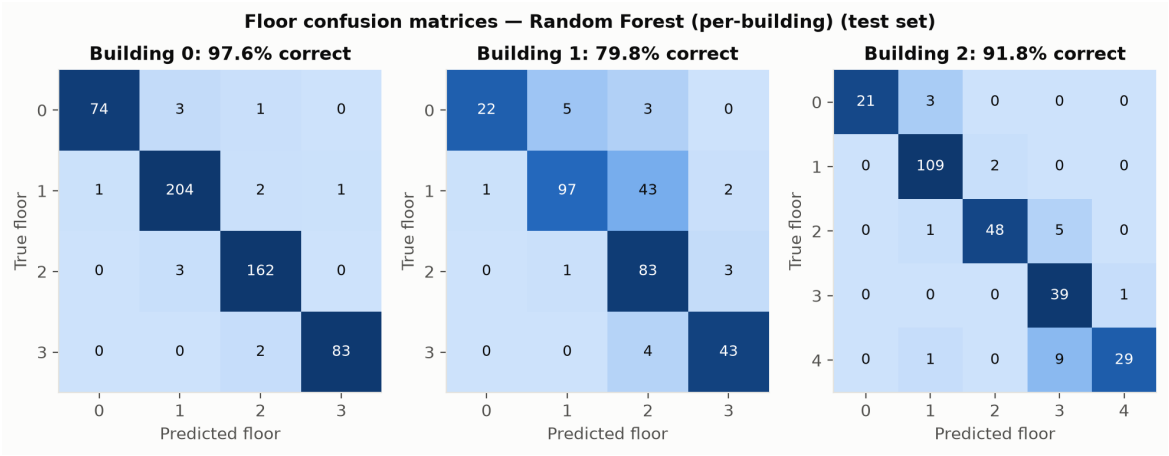


Fig. 12. Floor confusion matrices of the per-building Random Forest on the test set.



Fig. 13. Coordinate error map (per-building RF). Grey lines join the worst 10% of predictions to the true position.

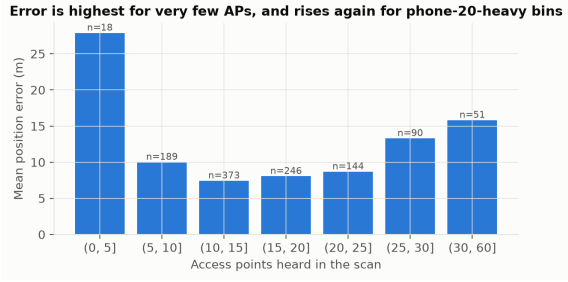


Fig. 14. Mean error by number of APs heard in the test scan.

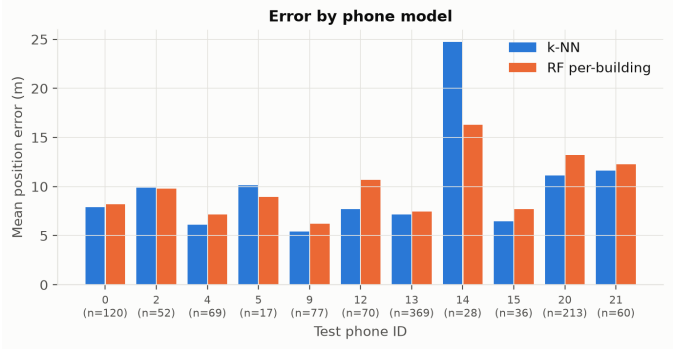


Fig. 15. Mean position error per test phone.

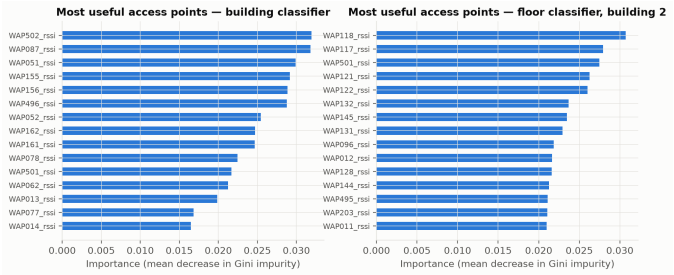


Fig. 16. Top-15 APs by Gini importance for the building and building-2 floor classifiers.

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